3.5.2 Roller Printing

It is a machine printing method, in which designs are printed on textile fabric by engraved rollers. This method of printing produces over 4000 yards of printed fabric per hour. The cylinder is made up of cast iron which acts as a printing table. The cylinder is covered with many layers of special fabric which has linen in warp and wool in weft for providing resiliency. It is covered with a layer of woollen blanket which provides the perfect surface for printing. The woollen blanket is covered with unbleached cotton cloth which will absorb excess dye. The last layer is the fabric which is to be printed.

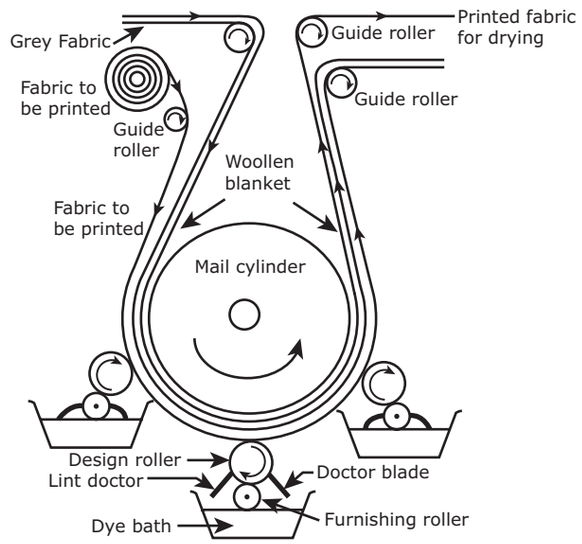


Figure 3.6 Schematic Diagram for Roller Printing

The design is transferred onto the cloth by engraved copper rollers. The design roller is arranged in such a way that the paste is applied on the roller and then transferred to the cloth. Number of engraved rollers rotate in contact with larger cylinder in rotation. The cloth is printed at the rate of 1000 to 4000 yards per hour. There are series of rollers each imprinting a different colour on the fabric. The sizes of the engraved cylinders depend on the pattern to be printed. There are different ways of engraving the roller such as hand engraving, machine or mill engraving and photographic engraving. The design roller rotates against the cloth and the design is imprinted. The fabric moves on to the second roller where the second colour is printed. The doctor blade placed in contact with design roller scrapes the excess dye from the surface of the design roller. The printed cloth is dried and then steamed to set the dye.

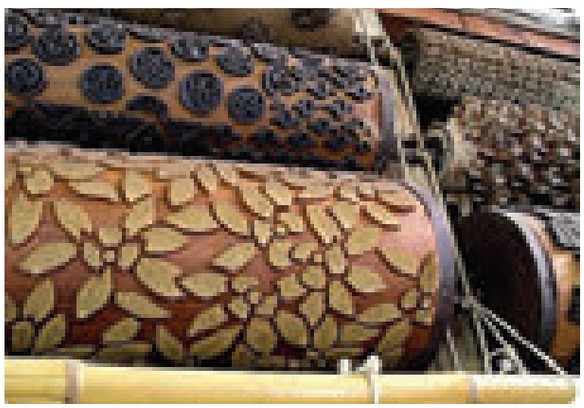


Figure 3.7 Engraved Rollers



Figure 3.8 Roller Printed Fabric

Advantages

- Superior to other methods for the production of fine and precise designs
- Production is faster and accurate

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